Logarithmic laws

1

a $\log_{10} 20$ **b** $\log_{10} 6$ **c** $\log_{10}\left(\frac{1}{4}\right)$ **d** $\log_5 198$

e $\log_{10}\left(\frac{35}{3}\right)$ **f** $\log_7 2$ **g** $3\log_{10} 18$ **h** $3\log_{10} 3$

i $2\log_5 2$ **j** $5\log_{10} 18$ **k** $\log_4 448$ **l** $\log_3 (10xy)$

2

a 1 **b** 3 **c** 3 **d** 2

e -2 **f** 2

3

a $7\log_4 x$ **b** $5\log(x+6)$ **c** $-\log(3x+7)$ **d** $\frac{2}{5}\log x$

e $5\log_6 y$ **f** $\frac{1}{2}\log_6 w$ **g** $\frac{1}{2}$ **h** $\frac{5}{4}$

i a **j** 10 **k** $\frac{3}{2}\log_5 x$ **l** $\frac{15}{4}$

4

a $\log_9 u + \log_9 v$ **b** $\log_5 9 - \log_5 7$

c $\log_2 5 + \log_2 x$ **d** $\log_{10} 2 - \log_{10} x$

e $\log m + \log m$ **f** $5(\log 3 + \log x)$

g $\log p + \log q - \log r$ **h** $-\log x - \log y$

i $-7(\log 5 + \log x)$ **j** $\log 5 + \frac{2}{3}\log x$

k $\frac{1}{3}(\log 14 + \log x)$ **l** $4\log c - \frac{1}{2}\log d$

5

a $7\log x$ **b** $\log(x^5 y^3)$ **c** $\log\left(\frac{x^8}{y^{\frac{1}{3}}}\right)$ **d** $\log\left(\frac{x^8}{y}\right)$

e $-35\log_{10} 5$ **f** $9\log_{10} 2$ **g** 2 **h** 2

6 $\log 2 + \log u - \log 3 - \log v$

7 $5\log x$

8

a $3\log u + 5\log v$ **b** $\frac{1}{3}\log v - \frac{1}{2}\log u$

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9

a 5 **b** 2 **c** 9 **d** -4

e 2 **f** 3 **g** 5 **h** 2

10 2.105

11

a 1.447 **b** 0.222 **c** 3.121 **d** -0.407

12

a 1.24 **b** 0.31 **c** 1.78 **d** -0.62

e 2.02

13

a 7.8 **b** 1.4 **c** 7.3 **d** -1.6

14 $\dfrac{\log_4 20}{\log_4 3}$

15

a $\dfrac{\log_{10} 16}{\log_{10} 4}$ **b** $\dfrac{\log_{10} 0.9}{\log_{10} 3}$ **c** $\dfrac{\log_{10} \sqrt{5}}{\log_{10} 3}$

16

a

i $\dfrac{\log_{10} 21}{\log_{10} 8}$ **ii** 1.46

b

i $\dfrac{\log_{10} \sqrt{5}}{\log_{10} 4}$ **ii** 0.58

c

i $\dfrac{\log_{10} 3^3}{\log_{10} 2}$ **ii** 4.75

d

i $\dfrac{\log_{10} 105}{\log_{10} \pi}$ **ii** 4.07

17

a $p + q$ **b** $p - q$ **c** $2q$ **d** $2p + 1$

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